

# Copper–Iron-Catalyzed Cross-Coupling of *ortho*-Substituted Sulfonamides with Sterically Hindered (Hetero)aryl Chlorides and Alkyl Halides: Synthesis of Sulfonamide Drugs

Keya Roy, Anay Saha, Sampa Mondal, and Laksmikanta Adak\*



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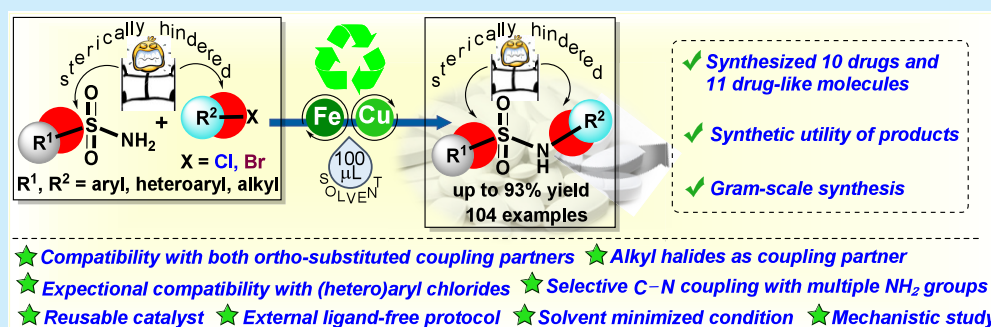
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**ABSTRACT:** We describe for the first time an effective copper–iron-catalyzed cross-coupling strategy that enables the reaction for a diverse array of *ortho*-substituted primary (hetero)aryl and aliphatic sulfonamides with challenging sterically hindered (hetero)aryl chlorides, bromides, and alkyl halides under solvent-minimized conditions. The developed methodology efficiently affords the corresponding products, *N*-aryl, *N*-heteroaryl, and *N*-alkyl sulfonamides bearing highly encumbered *ortho* substituents, in good to excellent yields with vast substrate scope (104 examples), tolerating several sensitive functionalities. In addition, selective C–N coupling of substrates containing multiple nucleophilic  $NH_2$  groups was effectively accomplished. This economically attractive protocol was effectively utilized for the synthesis of 10 commercially available sulfonamide-based drugs and 11 drug-like molecules. Notably, the methodology offers several other important advantages, including gram-scale synthesis, reusability of the catalyst, synthetic transformations of synthesized cross-coupled products, and elimination of conventional workup. Several spectroscopic experiments were conducted to identify the oxidation state involved in the active catalytic species. To explore the reaction mechanistic pathway, a radical clock experiment employing a radical probe and control reactions with radical scavengers were carried out.

*N*-Aryl and *N*-heteroaryl sulfonamides represent a vital class of compounds with broad applications in medicinal and agricultural chemistry.<sup>1</sup> Beyond these areas, sulfonamide derivatives also play important roles as organocatalysts,<sup>2</sup> chiral ligands,<sup>3</sup> and as key building blocks in various chemical transformations.<sup>4</sup> Consequently, the development of efficient and versatile synthetic strategies to construct diverse sulfonamide scaffolds is of considerable interest, particularly for accessing complex molecules relevant to drug discovery. However, the synthesis of structurally intricate sulfonamides, especially those bearing sterically hindered or heteroaryl substituents, remains a significant synthetic challenge. In particular, sterically congested *N*-aryl sulfonamides and *N*-heteroaryl sulfonamides are highly desirable motifs due to their prevalence in biologically active molecules.<sup>5</sup>

Traditional methods for synthesizing (hetero)aryl sulfonamides typically rely on the condensation of amine nucleophiles with reactive sulfonyl electrophiles such as arylsulfonyl halides, esters, azides, or hydrazides to form N–S bonds.<sup>6</sup> However,

these approaches often suffer from several drawbacks, including restricted diversity of substrates, poor tolerance toward functional groups, harsh reaction conditions, and side reactions. Moreover, concerns over the potential genotoxicity<sup>7</sup> of both amine nucleophiles and sulfonyl halides have prompted the development of alternative, safer synthetic strategies. In this context, C–N coupling reactions between aryl halides and sulfonamides catalyzed by palladium,<sup>8</sup> nickel,<sup>9</sup> and copper<sup>10</sup> in combination with ligands have emerged as a promising approach because of the commercial availability of both coupling partners.

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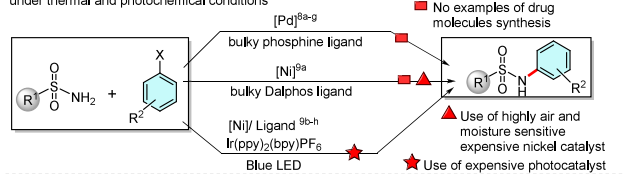


In recent decades, palladium-catalyzed coupling reactions have been widely employed to facilitate the coupling reaction of primary sulphonamides with aryl halides, typically in the presence of bulky phosphine ligands (Scheme 1a).<sup>8a–g</sup> These

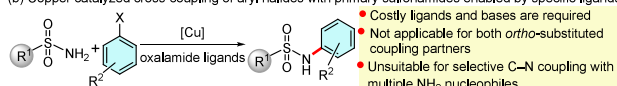
### Scheme 1. Transition-Metal-Catalyzed Cross-Coupling of Sulfonamides with Aryl and Alkyl Halides

**Previous Work:** Synthesis of *N*-arylated sulfonamides via cross-coupling reactions

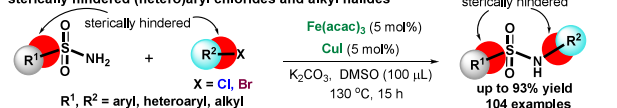
(a) Bulky ligand assisted palladium and nickel catalyzed cross-coupling of sulfonamides and aryl halides under thermal and photochemical conditions



(b) Copper catalyzed cross-coupling of aryl halides with primary sulfonamides enabled by specific ligands<sup>10a</sup>



(c) **This Work:** Iron-copper catalyzed cross-coupling of *ortho*-substituted sulfonamides with sterically hindered (hetero)aryl chlorides and alkyl halides



methodologies demonstrate excellent scope for (hetero)aryl bromides; however, their applicability to (hetero)aryl chlorides remains limited.<sup>8b–i</sup> Despite their usefulness, these protocols face several limitations, including the use of costly palladium catalysts and specialized ligands. Furthermore, the catalytic systems often exhibit poor reactivity with sterically hindered or heteroaromatic coupling partners.<sup>8a–d,f–i</sup>

Alternatively, cost-effective metals such as nickel with ligand have been explored for these transformations (Scheme 1a). These reactions are often conducted in the presence of expensive phosphine ligands,<sup>9a</sup> diamine-derived ligands,<sup>9b,e–g</sup> or ether-based ligands.<sup>9c,d,h</sup> While promising, these nickel-catalyzed methods are largely limited to aryl bromides, iodides, and a narrow subset of electron-deficient or electronically neutral aryl chlorides.<sup>9b,c,e–h</sup> Notably, examples involving electron-rich or sterically hindered (hetero)aryl chlorides remain scarce.

Recent advances in photochemical<sup>9b–d,f–h</sup> and electrochemical<sup>9e</sup> techniques have further expanded the utility of nickel catalysis for the formation of the C–N bond, particularly for reactions in-between aryl bromides and sulfonamides (Scheme 1a). These approaches often require the use of costly or toxic ligands, expensive photocatalysts, strong bases, or reactive alkoxide salts and additives, potentially limiting their scalability and widespread use.<sup>9</sup>

Researchers have also explored copper-based catalytic systems, often paired with electron-rich oxalamides<sup>10a</sup> or chelating 1,2-diamine ligands<sup>10b–e</sup> or polydentate chelating ligands,<sup>10f,g</sup> for *N*-arylation reactions of primary sulfonamides with aromatic halides. These ligands play a critical role in stabilizing the copper center and enhancing the catalytic activity in homogeneous reactions. However, copper-catalyzed methods have thus far been largely limited to (hetero)aryl iodides and bromides.<sup>10b–g</sup> These protocols often rely on high-cost and complex synthesis of some ligands, which limits their practicality. Ma and co-workers recently reported a noteworthy result through the development of a copper(I) oxide (Cu<sub>2</sub>O)-catalyzed system using oxalamide ligands for the coupling

reaction of primary sulfonamides with (hetero)aryl bromides and chlorides (Scheme 1b).<sup>10a</sup> However, the protocol still faces several major drawbacks, such as the requirement for costly electron-rich oxalamide ligands, the use of strongly reactive alkoxide bases. Moreover, the study did not include aliphatic halides, sterically hindered (hetero)aryl chlorides, and *ortho*-substituted sulfonamides as coupling partners, restricting its broader applicability. Recently, two reports have described ligand-free copper-catalyzed cross-coupling reactions of sulfonamides with aryl iodides and bromides.<sup>11</sup> However, these methods failed to afford any coupling product when applied to aryl chlorides or aliphatic reactants, and were also ineffective with heteroaromatic or sterically hindered coupling partners.

Pd, Ni, and Cu catalytic systems usually require the use of specific ligands and a strong base, which leads to the problems of functional group compatibility and low selectivity, particularly for sulphanilamide (selective C–N coupling with multiple nucleophilic NH<sub>2</sub> groups).<sup>8–10</sup> Furthermore, these cross-coupling methods often fail to couple sterically hindered (hetero)aryl chlorides with *ortho*-substituted sulfonamides. Notably, the coupling of alkyl halides with sulfonamides remains unexplored using Pd, Ni, or Cu catalysis, despite the importance of *N*-alkyl sulfonamides in drug development,<sup>5c</sup> highlighting a significant gap in current synthetic methodologies. Thus, the development of a general, efficient, and sustainable method to access structurally complex sulfonamides, including drug or drug-like scaffolds, remains a key goal in synthetic chemistry.

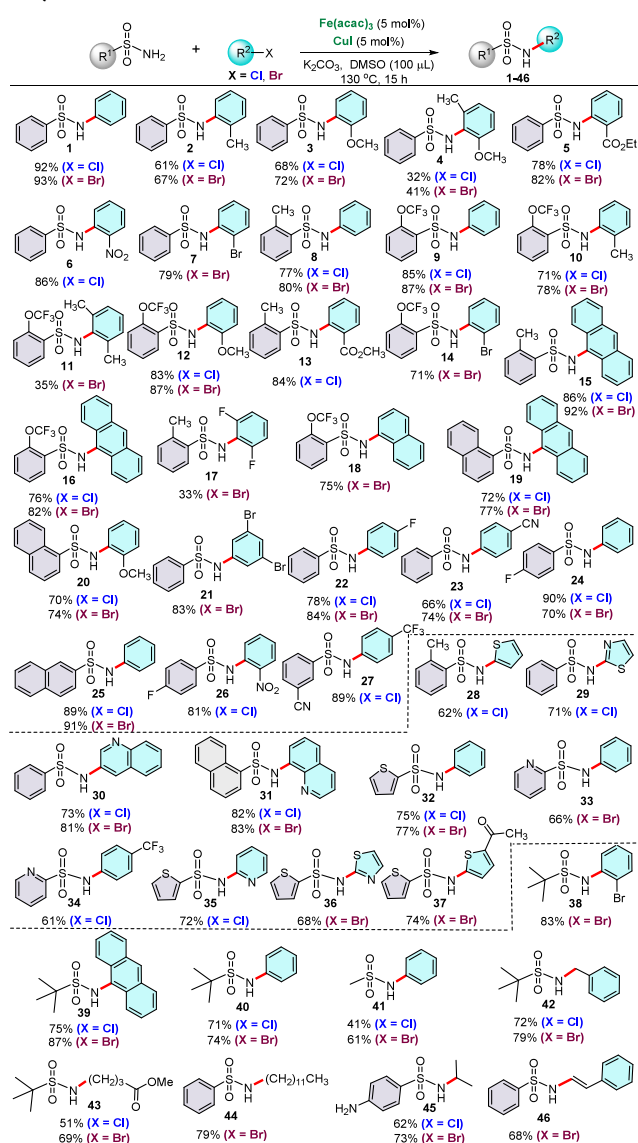
Recent developments in dual transition-metal catalysis have enabled effective approaches to cross-coupling reactions.<sup>12</sup> In particular, the synergistic combination of iron and copper catalysts has emerged as a promising strategy, leveraging the earth-abundance, low toxicity, and complementary reactivity of both metals, attributes that have shown great potential in cross-coupling chemistry. Recently, we have developed cost-effective, environmentally benign, and readily available bimetallic systems capable of efficiently facilitating C–N bond construction via cross-coupling between primary amides and aryl halides.<sup>13</sup>

Herein, we report for the first highly effective copper–iron-catalyzed cross-coupling reaction of a wide variety of heteroaryl and aryl chlorides and bromides, as well as alkyl halides, with diverse primary heteroaryl-, aryl-, and aliphatic sulfonamides under solvent-minimized conditions without using any externally added ligand (Scheme 1c). This bimetallic catalytic system proves particularly effective for the coupling of *ortho*-substituted primary sulfonamides with challenging, sterically hindered (hetero)aryl chlorides and bromides, substrates that have been notoriously difficult to engage using existing transition-metal catalysts (Pd, Ni, or Cu). The developed protocol is economically attractive and delivers the corresponding cross-coupling product, *N*-aryl, *N*-heteroaryl, and *N*-alkyl sulfonamides bearing highly encumbered *ortho* substituents in good to excellent yields.

To achieve the optimized reaction conditions, a series of experiments were conducted for the representative reaction between benzenesulfonamide and chlorobenzene.<sup>14a</sup>

To explore the reaction scope, a variety of aryl chlorides and bromides bearing diverse and challenging substituents, as well as several challenging sulfonamides, were initially investigated (Scheme 2). The coupling reaction between benzenesulfonamide and both chlorobenzene and bromobenzene afforded the targeted product **1** in excellent yield under standard reaction conditions. Subsequently, we focused on exploring the cross-coupling reaction of sterically hindered partners, which

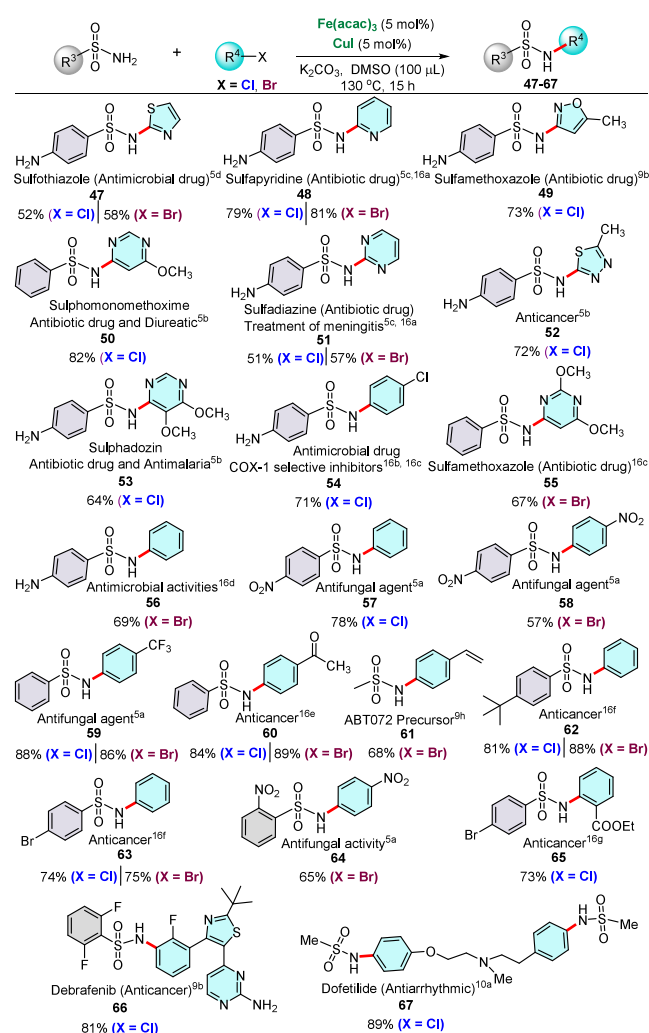
## Scheme 2. Copper–Iron-Catalyzed Cross-Coupling of (Hetero)aryl or Aliphatic Sulfonamides with (Hetero)aryl or Alkyl Chlorides and Bromides<sup>a</sup>



<sup>a</sup>Reaction conditions: (hetero)aryl or alkyl sulfonamides (0.5 mmol), (hetero)aryl or alkyl halides (0.75 mmol),  $Fe(acac)_3$  (5 mol %),  $CuI$  (5 mol %),  $K_2CO_3$  (0.75 mmol), DMSO (100  $\mu L$  for 0.5 mmol scale of reaction) under  $N_2$  atmosphere. Isolated yields are reported.

successfully provided the desired products (2–9) in high yields.<sup>15</sup> The reaction of 2-chloroanisole with benzenesulfonamide using a copper catalyst in the presence of a ligand was previously attempted, but resulted in a trace amount of the product 3.<sup>10a</sup> This observation showcases the efficiency of our established catalytic system toward deactivated aryl chlorides. We extended our protocol to more challenging conditions by introducing steric bulk to both coupling partners. We were pleased to find that our developed methodology performed with high success in reactions between coupling partners possessing sterically hindered *ortho* substituents, yielding the products 10–20 in high yield, highlighting the exceptional compatibility of our method with steric challenges. Notably, several aryl chlorides and bromides bearing substituents at the *ortho*, *meta*, and *para* positions successfully underwent coupling

## Scheme 3. Synthesis of Marketed Drugs and Drug-like Molecules via Copper–Iron-Catalyzed Cross-Coupling Reaction<sup>a</sup>



<sup>a</sup>Reaction conditions: aryl or alkyl sulfonamide (0.5 mmol), (hetero)aryl chloride or bromide (0.75 mmol),  $Fe(acac)_3$  (5 mol %),  $CuI$  (5 mol %),  $K_2CO_3$  (0.75 mmol), DMSO (100  $\mu L$  for 0.5 mmol scale of reaction), under  $N_2$  atmosphere. Isolated yields are reported.

reactions with *meta*- and *para*-substituted arylsulfonamides as well as polyaromatic sulfonamides, yielding products 21–27 in good to excellent yields. It is notable that 1,2-dibromobenzene and 1,3,5-tribromobenzene underwent selective cross-coupling with aryl sulfonamides, affording products 7, 14, and 21 in excellent yields. The retained bromide groups in these products indicate their suitability for subsequent cross-coupling processes, thereby broadening the possibilities for further chemical modifications.

The coupling reaction was investigated with diverse heteroaromatic structures for both coupling partners (Scheme 2). (Hetero)aryl chlorides and bromides reacted successfully with (hetero)arylsulfonamides, providing *N*-heteroaryl sulfonamides 28–34 with 61–83% yield. The protocol was not restricted to single heteroaromatic coupling partners, as the optimal reaction condition was efficiently applied for the coupling reaction of heteroaromatic primary sulfonamides with heteroaryl chlorides and bromides, providing highly

enriched heteroatomic structured secondary sulfonamides **35**–**37** with moderate to good yield.

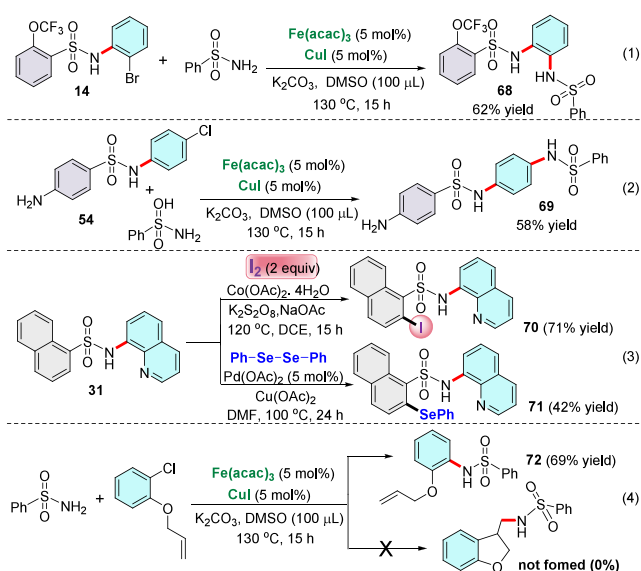
The coupling reactions are not limited to aromatic or heteroaromatic substrates; they also proceeded effectively with sterically hindered aliphatic primary sulfonamides and challenging alkyl chlorides and bromides, as well as styryl bromide, to produce coupling products **38**–**46** in high yields (Scheme 2). These results highlight the wide applicability and robust nature of this methodology.

Finally, to demonstrate the practicability of this novel copper–iron-catalyzed method and its potential application for the synthesis of commercially available drugs and drug-like molecules, we prepared a range of pharmaceutically established secondary sulfonamide drugs **5a–c,9b,h,10a,16** (Scheme 3).

In this study, it was observed that 4-amino benzenesulfonamide underwent a highly selective C–N cross-coupling reaction, leaving the primary amine group intact. This selectivity toward C–N coupling in the presence of multiple nucleophilic NH<sub>2</sub> groups enabled the efficient synthesis of antibiotic, antimicrobial, and antimalarial drug molecules **47**–**56** with good yields. Aryl chlorides and bromides, bearing trifluoromethyl and other sensitive functional groups including nitro, acyl, ester, and vinyl, are well tolerated, resulting in biologically and medically demanding secondary sulfonamides (**57**–**65**) with yields of up to 89%. Furthermore, our optimal cross-coupling strategy was effectively employed in the synthesis of Dabrafenib (**66**), an anticancer drug, achieving an 81% yield, and Dofetilide (**67**), an antiarrhythmic drug, which was obtained in 89% yield. These results highlight the practical utility of our methodology in pharmaceutical synthesis.

To showcase the practical utility of the cross-coupling reaction, a synthesis on a gram scale was conducted under the optimized reaction conditions, affording the desired secondary sulfonamide product **13** in a comparable yield within the same time period relative to the lower millimolar reaction scale.<sup>14b</sup>

To assess the practical applicability of the protocol, we carried out several synthetic transformations using our synthesized cross-coupled products **1**,<sup>14c</sup> **14**, **31**, and **54**. The products **14** and **54** (antimicrobial drug) underwent further cross-coupling with benzenesulfonamides under the optimized reaction conditions, affording the extended cross-coupled products **68** and **69** in 62% and 58% yield, respectively (eqs 1 and 2). Moreover, we employed product **31** in C–H functionalization reactions where the quinoline moiety acts as a directing group. Under cobalt catalysis, compound **31** underwent efficient C(sp<sup>2</sup>)–H iodination, while palladium catalysis facilitated C(sp<sup>2</sup>)–H selenylation, yielding products **70** and **71**, respectively, in 71% and 42% yields (eq 3). The catalyst was successfully reused for up to five consecutive cycles without significant loss of catalytic activity.<sup>14d</sup>



To investigate the mechanism of the cross-coupling reaction, preliminary mechanistic studies were conducted. As an initial approach, a radical clock experiment was performed under standard reaction conditions employing 1-(allyloxy)-2-chlorobenzene as a radical probe (eq 4). This experiment yielded the direct cross-coupling product **72** without generating a ring-closed cross-coupling product, indicating that radical intermediates are not generated in the reaction. Subsequently, the reaction was conducted in the presence of a few radical scavengers, and a noteworthy amount of product **1**, was still isolated, further supporting the conclusion that the reaction proceeds through a nonradical pathway.<sup>14e</sup>

A series of experiments were carried out to identify the oxidation state of the active catalytic species participating in the mechanistic cycle and to explore the reaction mechanism. X-ray photoelectron spectroscopy (XPS) was utilized to perform the high-resolution Fe 2p spectrum of the sample.<sup>14f</sup> This experiment confirms the presence of Fe(II) and Fe(III) oxidation states in the reaction mixture.<sup>13a,17</sup> Furthermore, the experiment also suggests that Cu is present in the +1 oxidation state within the active catalytic species.<sup>14g,13a,18</sup>

UV spectroscopy was analyzed which supports the presence of Fe(II), Fe(III), and Cu(I) species in the reaction mixture.<sup>14h</sup> Powder X-ray diffraction (PXRD) analysis of the residual powder, collected after extracting the organic compounds,<sup>14i</sup> indicated the existence of both Fe(II) and Fe(III) species, along with Cu(I).<sup>13a,19</sup>

Based on mechanistic studies and prior literature reports,<sup>10d–f,20</sup> a plausible mechanistic pathway is proposed in Scheme S2. The proposed reaction mechanism proceeds through three elementary steps: (i) The aryl halide undergoes oxidative addition to the catalytically active copper(I) species **A**, generating intermediate **B**.<sup>10d–f,20</sup> The sulfonamide substrate or acetylacetonate (acac) might coordinate with the active catalytic species, potentially assisting in the oxidative addition process. (ii) transmetalation of intermediate **B** with an iron-amide species to yield intermediate **C**;<sup>21</sup> and (iii) reductive elimination from intermediate **C** to produce the cross-coupling product and to regenerate the involved active copper(I) species **A**.

In summary, we report the first efficient copper–iron-catalyzed cross-coupling reaction of sterically hindered (hetero)aryl chlorides and bromides with *ortho*-substituted primary sulfonamides under solvent-minimized conditions and

in the absence of any externally added ligand. This method was successfully extended to a broad range of aryl, heteroaryl, and alkyl chlorides and bromides, as well as various primary heteroaryl-, aryl-, and aliphatic sulfonamides. It demonstrates a wide scope of substrates with 104 examples and shows excellent compatibility with various sensitive functional groups, including  $-\text{COCH}_3$ ,  $-\text{CO}_2\text{Me}$ ,  $-\text{CO}_2\text{Et}$ ,  $-\text{CN}$ ,  $-\text{NO}_2$ ,  $-\text{NH}_2$ ,  $-\text{Br}$ ,  $-\text{Cl}$ ,  $-\text{F}$ ,  $-\text{CF}_3$ ,  $-\text{OCF}_3$ ,  $-\text{OCH}_3$ ,  $-\text{CH}_2\text{Ph}$ , vinyl, and heterocycles. Notably, this approach allows for selective C–N cross-coupling reaction despite the presence of multiple nucleophilic  $\text{NH}_2$  groups, preserving the primary amine functionality. Additionally, the method has been effectively utilized in the synthesis of 10 commercially available sulfonamide-based drugs and 11 pharmaceutically important scaffolds, highlighting its significance in medicinal chemistry. Several synthetic transformations, such as further cross-coupling and C–H functionalization, were performed using our synthesized cross-coupled products. The protocol is also applied to gram-scale synthesis, demonstrating its practicality for larger-scale applications. Moreover, this protocol avoids the need for conventional workup procedures, enhancing overall efficiency and simplicity. Importantly, the catalyst employed is environmentally benign, cost-effective, and demonstrates reusability for at least four reaction cycles. The reaction mechanism for the cross-coupling was also investigated.

## ■ ASSOCIATED CONTENT

### Data Availability Statement

The data underlying this study are available in the published article and its [Supporting Information](#).

### SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.orglett.5c04324>.

Experimental details, procedures, compound characterization data, and NMR spectra for all the synthesized compounds (PDF)

## ■ AUTHOR INFORMATION

### Corresponding Author

Laksmikanta Adak – Department of Chemistry, Indian Institute of Engineering Science and Technology, Howrah 711103, India; [orcid.org/0000-0002-8544-7615](https://orcid.org/0000-0002-8544-7615); Email: [ladak@chem.iiests.ac.in](mailto:ladak@chem.iiests.ac.in)

### Authors

Keya Roy – Department of Chemistry, Indian Institute of Engineering Science and Technology, Howrah 711103, India; [orcid.org/0000-0003-4984-1882](https://orcid.org/0000-0003-4984-1882)

Anay Saha – Department of Chemistry, Indian Institute of Engineering Science and Technology, Howrah 711103, India; [orcid.org/0009-0004-2423-1951](https://orcid.org/0009-0004-2423-1951)

Sampa Mondal – Department of Chemistry, Indian Institute of Engineering Science and Technology, Howrah 711103, India

Complete contact information is available at:

<https://pubs.acs.org/doi/10.1021/acs.orglett.5c04324>

### Notes

The authors declare no competing financial interest.

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tion Cascade Involving Ortho-Azidobenzaldehydes and  $\beta$ -Ketosulfonamides and Sulfones. *Beilstein J. Org. Chem.* **2023**, *19*, 800–807. (f) See [Figure S1](#) for examples of some therapeutic agents and drug molecules that contain N-aryl sulfonamide groups.

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(14) (a) See [Table S1](#) for the optimization study of reaction conditions. Experimental procedure: A mixture of benzenesulfonamide (78 mg, 0.5 mmol), chlorobenzene (84 mg, 0.75 mmol), K<sub>2</sub>CO<sub>3</sub> (104 mg, 0.75 mmol), Fe(acac)<sub>3</sub> (9 mg, 5 mol%), CuI (5 mg, 5 mol%), and DMSO (100  $\mu$ L per 0.5 mol reaction scale) was heated at 130 °C under a nitrogen atmosphere for 15 h in an oil bath. The reaction mixture was then allowed to cool and filtered through a short column using ethyl acetate. The filtrate was concentrated under reduced pressure, and the resulting crude product was purified by column chromatography over silica gel (15% ethyl acetate in hexane) to afford the pure product as a white solid (107 mg, 92%). (b) See [eq S1](#) for the gram-scale synthesis. (c) See [eq S2](#) for the synthesis of tertiary sulfonamide **73** by further

reaction of the C–N coupling product **1**. (d) See Figure S2 and Scheme S1 for details on the reusability and recovery procedure of the copper–iron catalyst. (e) See eq S3 for the yield of product **1** obtained under the optimized conditions in the presence of 2,2,6,6-tetramethylpiperidine-1-oxyl, benzyl alcohol, and butylated hydroxytoluene. (f) See Figure S3a for XPS study of the reaction mixture for the presence of Fe(II) species. (g) See Figure S3b for the XPS study for the presence of Cu(I) as an active catalytic species. (h) See Figure S4a for UV spectra of the reaction mixture. (i) See Figure S4b for PXRD analysis of Fe(II), Fe(III), and Cu(I) present in the residual powder left after the extraction of organic compounds.

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